

Experiment 3:

Aim: To analyse the performance of a multi class confusion matrix by using choosen database with python code

Algorithm:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
from sklearn.metrics import accuracy_score, classification_report
import matplotlib.pyplot as plt

# Load the Iris dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Train the Decision Tree Classifier
model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)

# Predict the test data
y_pred = model.predict(X_test)
```



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# Predict the test data
y_pred = model.predict(X_test)

# Generate the confusion matrix
cm = confusion_matrix(y_test, y_pred)

# Print confusion matrix
print("Confusion Matrix:")
print(cm)

# Display confusion matrix
disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=iris.target_names
)

disp.plot(cmap="Blues")
plt.title("Multi-Class Confusion Matrix")
plt.show()

# Print Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy: {:.2f}%".format(accuracy * 100))

# Print Classification Report
print("\nClassification Report:")
print(classification_report(y_test, y_pred,
                           target_names=iris.target_names))
```

OUTPUT:



